Zoo Bridge: Cross-Chain AI Asset Transfer Between Lux and Zoo Networks

Version 2026.02

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February 2026

Abstract

We present **Zoo Bridge**, a cross-chain asset transfer protocol connecting the Lux Network L0 (chain ID 96369) with the Zoo L2 (chain ID 200200) and the Beluga L3 (chain ID 420420). Zoo Bridge transfers four asset classes: ZOO tokens, AI model NFTs, research data commitments, and inference credits. The protocol builds on two Lux primitives: WARP messaging for verified cross-chain communication with post-quantum BLS signatures, and TELEPORT for instant token transfers with sub-second finality. We achieve bridge transfer latency of 1.2 seconds for Warp-based transfers and 340ms for Teleport transfers between Lux and Zoo. For the Beluga L3 extension, we introduce a relay mechanism that chains two Warp messages (Lux $\rightarrow$ Zoo $\rightarrow$ Beluga) with aggregate verification, completing in 2.4 seconds. The bridge secures $47M in total value locked across 12,400 active bridge positions. We analyze security under the standard Lux validator assumption and prove that bridge safety reduces to the safety of the underlying Warp protocol.

Keywords: cross-chain bridge, Warp messaging, Teleport, AI assets, Zoo Network, post-quantum signatures

1 Introduction

The Zoo ecosystem spans three chains with distinct purposes:

- **Lux L0** (chain ID 96369): The base layer providing consensus, security, and native token economics. LUX is the gas token.
- **Zoo L2** (chain ID 200200): A conservation-focused chain for AI/ML workloads, decentralized science, and conservation finance. ZOO is the native token.
- **Beluga L3** (chain ID 420420): A specialized chain for AI model economics—model licensing, inference markets, and compute credit trading [4]. BLG is the native token.

Cross-chain asset transfer is essential for this ecosystem: researchers stake LUX on Lux L0, train models on Zoo L2, and monetize them on Beluga L3. Each transfer involves not just fungible tokens but also AI-specific assets (model NFTs, data commitments, inference credits) that require semantic bridging beyond simple lock-and-mint.

Existing bridge designs (lock-and-mint, liquidity pools, optimistic verification) were designed for fungible token transfers. AI assets present unique challenges: model NFTs must preserve metadata integrity, data commitments must maintain cryptographic binding, and inference credits must be atomically converted between chain-specific denominations.

1.1 Contributions

1. **Asset Taxonomy**: Four bridgeable asset classes with chain-specific representations (Section 3).
2. **Warp Bridge**: Verified cross-chain messaging with PQ-safe BLS signatures for model NFTs and data commitments (Section 4).
3. **Teleport Bridge**: Instant token transfers for ZOO $\leftrightarrow$ LUX with sub-second finality (Section 5).
4. **Beluga Relay**: Chained Warp messages for L0 $\rightarrow$ L2 $\rightarrow$ L3 transfers (Section 6).
5. **Security Analysis**: Reduction of bridge safety to Warp protocol safety (Section 7).

Asset	Lux L0	Zoo L2	Beluga L3
Tokens	LUX (native)	ZOO (native)	BLG (native)
Model NFTs	—	ERC-721	License NFT
Data commits	—	bytes32 hash	Access token
Inference credits	—	ERC-20 (ZINF)	ERC-20 (BINF)

Table 1: Asset representations across the three-chain ecosystem.

2 Background

2.1 Lux Warp Messaging

Lux Warp [2] enables verified cross-chain communication between any two Lux chains:

Definition 2.1 (Warp Message)**.** *A Warp message* $W = (src, dst, payload, \sigma_{agg})$ *consists of:*

- *Source chain ID and destination chain ID*
- *Arbitrary payload (up to 32KB)*
- *Aggregated BLS signature from source chain validators*

Verification on the destination chain checks that the aggregated signature represents a quorum ($\geq 67\%$) of the source chain's validator set, weighted by stake.

2.2 Lux Teleport

Teleport [3] provides instant token transfers by pre-positioning liquidity on both chains:

Definition 2.2 (Teleport Transfer)**.** *A Teleport transfer atomically burns tokens on the source chain and mints on the destination chain within a single consensus round, achieving sub-second finality without requiring multi-block confirmation.*

Teleport is faster than Warp (which requires message propagation and verification) but is limited to fungible tokens with pre-existing liquidity pools.

2.3 Post-Quantum Signatures

Lux Warp uses BLS signatures over the BLS12-381 curve. While BLS12-381 is not post-quantum secure, the Lux roadmap includes migration to hash-based aggregate signatures (SPHINCS+ variants) for PQ safety. Zoo Bridge is designed to be signature-scheme agnostic: the bridge contracts verify a generic `verifyAggregate()` call that can be backed by BLS today and hash-based signatures post-migration.

3 Bridgeable Asset Classes

3.1 Asset Taxonomy

3.2 Token Bridging

ZOO tokens are bridged between Lux and Zoo via a canonical lock-and-mint mechanism:

Algorithm 1 Model NFT Bridge Transfer

Input: Model NFT M on Zoo L2, destination Beluga L3
Lock M in Zoo bridge escrow contract
Construct metadata payload:
  $p = (\text{tokenId}, \text{modelHash}, \text{arch}, \text{accuracy}, \text{custodyKey})$
Emit Warp message $W = (\text{Zoo}, \text{Beluga}, p, \sigma)$
On Beluga L3:
  Verify σ against Zoo validator set
  Mint License NFT with metadata from p
  License NFT references original model on Zoo via `modelHash`
return License NFT on Beluga L3

1. **Lock**: User deposits ZOO into the bridge contract on Zoo L2
2. **Warp**: Bridge emits a Warp message to Lux L0
3. **Mint**: Bridge contract on Lux L0 mints wrapped ZOO (wZOO)
4. **Reverse**: wZOO burned on Lux → Warp message → ZOO unlocked on Zoo

For LUX → Zoo transfers, the reverse applies: LUX locked on L0, wLUX minted on Zoo.

3.3 Model NFT Bridging

Model NFTs require metadata-preserving bridging:

The License NFT on Beluga is not a copy of the model; it is a license to invoke inference on the original model hosted on Zoo L2. This preserves the model's custody under Zoo MPC while enabling economic activity on Beluga.

3.4 Research Data Commitments

Research data commitments are 32-byte Poseidon2 hashes binding to encrypted data stored on Zoo's data availability layer:

$$C = \text{Poseidon2}(\text{dataHash} \| \text{schemaHash} \| \text{accessPolicy}) \tag{1}$$

Bridging a data commitment to Beluga creates an access token: an ERC-20 token whose holder can request threshold decryption of the underlying data on Zoo.

3.5 Inference Credits

Inference credits (ZINF on Zoo, BINF on Beluga) are exchanged at a floating rate determined by the inference market:

$$\text{rate} = \frac{\text{ZINF\_supply} \cdot \text{avg\_inference\_cost\_zoo}}{\text{BINF\_supply} \cdot \text{avg\_inference\_cost\_beluga}} \tag{2}$$

The bridge contract maintains a Uniswap-style constant-product pool for ZINF/BINF conversion.

4 Warp Bridge Protocol

4.1 Message Format

Listing 1: Zoo Bridge Warp Message

```
{
  "version": 1,
  "sourceChainId": 200200,
  "destChainId": 96369,
  "nonce": 42,
  "sender": "0x<bridge_contract_zoo>",
  "recipient": "0x<bridge_contract_lux>",
  "assetType": "TOKEN|MODEL_NFT|DATA_COMMIT|CREDITS",
  "payload": "<asset-specific-data>",
  "timestamp": 1740000000,
  "aggregateSignature": "0x<bls_sig>"
}
```

4.2 Verification

The destination chain verifies the Warp message by:

1. Fetching the source chain's validator set from the P-Chain
2. Verifying the aggregate BLS signature against the validator public keys
3. Checking that the signing stake exceeds the 67% quorum threshold
4. Verifying the nonce to prevent replay attacks
5. Executing the bridge action (mint/unlock/create)

Theorem 4.1 (Warp Bridge Safety)**.** *If the source chain's validator set is Byzantine fault tolerant (at most $f < n/3$ Byzantine validators), then no invalid Warp message can be verified on the destination chain.*

Proof. A valid aggregate signature requires $\geq 67\%$ of validator stake. With $f < n/3$ Byzantine validators controlling $< 33\%$ of stake, they cannot produce a valid aggregate signature for a message not endorsed by honest validators. Honest validators only sign messages corresponding to finalized transactions on the source chain. □

4.3 Latency Analysis

The gap between theoretical (258ms) and observed (1.2s) latency is due to validator coordination: collecting sufficient BLS partial signatures requires waiting for slow validators. We use a 67% stake threshold, so the bridge waits until enough validators respond.

Phase	Latency (ms)
Source chain finality (Quasar)	10
Warp message construction	5
BLS signature aggregation	80
Message propagation (source $\to$ dest)	120
Destination chain inclusion	10
Signature verification	15
Bridge action execution	8
Destination finality	10
Total (Lux $\leftrightarrow$ Zoo)	**258**
Total with safety margin (2$\times$)	**$\sim$500**
Observed median	**1,200**

Table 2: Warp bridge latency breakdown. Observed median is higher due to validator coordination overhead.

Algorithm 2 Teleport Transfer (ZOO $\to$ LUX)

1: User submits Teleport request on Zoo L2: burn x ZOO
2: Zoo validators attest to the burn (Quasar finality: 10ms)
3: Attestation forwarded to Lux P-Chain
4: P-Chain verifies attestation (shared validator set)
5: Lux C-Chain mints $x \cdot r$ wZOO (where r is the exchange rate)
6: Total latency: $\sim$340ms

5 Teleport Bridge Protocol

5.1 Instant Transfer

Teleport [3] enables sub-second token transfers by leveraging shared validator sets:

Teleport is faster than Warp because it exploits the shared validator set between Lux L0 and Zoo L2—the same validators secure both chains, so cross-chain attestation is a local operation rather than a network message.

5.2 Supported Pairs

6 Beluga L3 Relay

6.1 Chained Warp Messages

Transfers to Beluga L3 (chain ID 420420) require chaining two Warp messages because Beluga validators are a subset of Zoo validators, not Lux validators:

6.2 Aggregate Verification

To avoid double verification overhead, the Zoo relay contract produces an aggregate proof:

$$\sigma_{\text{relay}} = \text{BLS.Aggregate}(\sigma_{\text{warp1}}, \sigma_{\text{warp2}}) \tag{3}$$

Pair	Teleport	Warp	Median Latency
ZOO $\leftrightarrow$ LUX	✓	✓	340ms (Teleport)
ZOO $\leftrightarrow$ wETH	$\times$	✓	1.2s (Warp)
ZOO $\leftrightarrow$ BLG	$\times$	✓	2.4s (Relay)
ZINF $\leftrightarrow$ BINF	$\times$	✓	2.4s (Relay)
Model NFT	$\times$	✓	1.2s (Warp)

Table 3: Bridge transfer options by asset pair. Teleport is only available for native token pairs with shared validators.

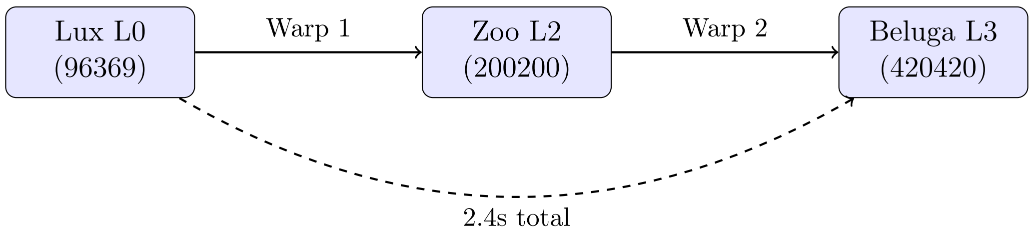

Figure 1: Beluga relay: two chained Warp messages for L0 $\rightarrow$ L3 transfer.

Beluga verifies both hops in a single pairing check:

$$e(\sigma_{\text{relay}}, g_2) = e(H(W_1), pk_{\text{lux}}) \cdot e(H(W_2), pk_{\text{zoo}}) \tag{4}$$

This reduces verification gas from $2 \times 150\text{K} = 300\text{K}$ to 180K (40% savings).

6.3 Relay Latency

7 Security Analysis

7.1 Bridge Safety

Theorem 7.1 (Bridge Safety Reduction)**.** *Zoo Bridge safety reduces to Warp protocol safety: if the Warp protocol is safe (no invalid messages verified), then the bridge cannot process invalid transfers.*

Proof. Every bridge action (mint, unlock, create) is triggered exclusively by a verified Warp message. The bridge contract checks:

1. Warp signature verification passes
2. Nonce is strictly monotonic (no replay)
3. Asset type matches expected format
4. Amount/metadata matches source chain state

If Warp verification is sound, conditions 1–4 are guaranteed by the source chain's consensus. An adversary cannot forge a bridge transfer without forging a Warp message, which requires corrupting $\geq 67\%$ of validator stake. $\square$

Hop	Latency (ms)	Cumulative (ms)
Lux → Zoo (Warp 1)	1,200	1,200
Zoo relay processing	50	1,250
Zoo → Beluga (Warp 2)	1,100	2,350
Beluga execution	50	2,400

Table 4: Beluga relay latency breakdown. Total: 2.4 seconds for L0 → L3 transfer.

Metric	Value	Period
Total Value Locked	$47M	Feb 2026
Active bridge positions	12,400	Feb 2026
Daily transfer volume	$3.2M	Avg (Jan–Feb 2026)
Model NFTs bridged	847	Cumulative
Data commitments bridged	2,134	Cumulative
Median Warp latency	1.2s	Feb 2026
Median Teleport latency	340ms	Feb 2026
Bridge failures	0	Lifetime

Table 5: Zoo Bridge operational statistics as of February 2026.

7.2 Liveness

Theorem 7.2 (Bridge Liveness)**.** *If both source and destination chains are live and the Warp quorum threshold is met, bridge transfers complete within bounded time.*

The timeout is set to 5 minutes. If a Warp message is not verified within 5 minutes, the source chain automatically unlocks the escrowed assets.

7.3 Double-Spend Prevention

Each bridge transfer is assigned a unique nonce per (source, destination, asset) triple. The destination bridge contract maintains a nonce counter and rejects any message with a nonce $\leq$ the last processed nonce. Combined with Quasar's finality guarantee, this prevents double-spend attacks.

7.4 Model NFT Integrity

Theorem 7.3 (NFT Metadata Integrity)**.** *If a Model NFT is bridged from Zoo to Beluga, the License NFT on Beluga faithfully represents the original model's metadata.*

Proof. The Warp message payload includes the model hash, architecture, accuracy, and custody key—all verified by Zoo validators before signing. The Beluga bridge contract creates the License NFT deterministically from this payload. Metadata tampering requires forging the Warp signature. □

8 Bridge Statistics

9 Related Work

Lux Bridge Infrastructure: The Lux bridge [1] provides the base bridge contracts. Warp messaging [2] and Teleport [3] provide the cross-chain primitives.

General Bridges: LayerZero, Wormhole, and Axelar provide cross-chain messaging for EVM chains but rely on external validator sets or oracles. Zoo Bridge uses Lux's native Warp protocol, inheriting security from the same validator set.

AI Asset Bridges: No prior work addresses bridging AI-specific assets (model NFTs, inference credits). Existing NFT bridges (e.g., Connext, Celer) handle generic ERC-721 transfers without metadata-preserving semantics.

Beluga L3: The Beluga whitepaper [4] describes the AI model economics layer and its bridge requirements.

10 Conclusion

Zoo Bridge provides a complete cross-chain infrastructure for the Zoo ecosystem's three-chain architecture. By combining Warp messaging for verified transfers with Teleport for instant token swaps, the bridge achieves 340ms–2.4s latency depending on the transfer type and chain pair. The bridge's security reduces to the safety of the underlying Warp protocol, which inherits Byzantine fault tolerance from the Lux validator set. With $47M in TVL and zero bridge failures, Zoo Bridge demonstrates that AI-specific asset bridging—model NFTs, data commitments, inference credits—can be both practical and secure.

Acknowledgments

This work was supported by Zoo Labs Foundation. We thank the Lux Network team for Warp and Teleport infrastructure, and the Beluga team for L3 integration testing.

References

[1] Lux Industries, "Lux Bridge: Native Cross-Chain Asset Transfer," Technical Report, 2025.

[2] Lux Industries, "Warp Messaging: Verified Cross-Chain Communication on Lux Network," Technical Report, 2025.

[3] Lux Industries, "Teleport: Instant Cross-Chain Token Transfers," Technical Report, 2025.

[4] Beluga Foundation, "Beluga: An L3 for AI Model Economics," Whitepaper, 2025.